



Calculation Sheet

Calculation header

Identifier

*New calculation (1)***Medium selection and state**

Medium

Water

State

*Liquid***Operating data**☐ Safety-related application

	Maximum flow	Mean flow	Minimum flow	
t1	60.0	☒ 0.0	☒ 0.0	°C
p1	10.0	0.0	0.0	MPa(g)
☐ p2	4.0			MPa(g)
☒ Δp	6.0	☒ 0.0	☒ 0.0	MPa
Cv	74.159	☒	☒	GPM(US)
☐ qm	493,760.0			kg/h
☒ qv	500.0	☒ 0.0	☒ 0.0	m³/h
LpAe	☒ 109.0			dB(A)
	☒ Incipient cavitation			

Fluid operating data

	Maximum flow	Mean flow	Minimum flow	
ρ1	987.52	☒	☒	kg/m³
ρ2	984.32			kg/m³
☐ η1	0.46867	☒	☒	mPa s
☒ ν1	0.47459			mm²/s
pv1	0.19946	☒	☒	bar(a)
pv2	0.21085			bar(a)
tv1	311.74	99.974	99.974	°C
cF1	1,569.4			m/s

Pipe downstream of valve

Size class downstream of valve

NPS2

2"

Schedule downstream

SCH2

STD



Calculation Sheet

Valve configuration

Valve type	<i>Straight globe valve</i>
Trim type	<i>Contoured plug</i>
Flow direction	<i>FTC</i>
Valve performance class	<i>Multi stage valve</i>
Protection	<i>Non-hardened</i>
Low-noise design	<i>n.a.</i>

Valve data

Basic characteristic		<i>Linear</i>	
☐ Lift stopper			
Selected valve size	NPS	<i>2"</i>	
Pressure class	class	<i>class 600</i>	
Maximum operating pressure	p,max	<i>105.0</i>	bar(a)
Valve modifier (Cv/Cv100 = 0.75)	FL ²	<i>0.894</i>	-
Valve modifier (Cv/Cv100 = 0.75)	Kc	<i>0.846</i>	-
Cavitation modifier (Cv/Cv100 = 0.75)	xFz	<i>0.527</i>	-
Valve style modifier (Cv/Cv100 = 1)	Fd	<i>0.41</i>	-

Load-dependent values

		Maximum flow	Mean flow	Minimum flow	
u2	!	<i>68.748</i>			m/s
LpAe	!	<i>109.0</i>			dB(A)
Ri	!	<i>7.5</i>			-

Noise calculation

Calculation standard (noise, liquid)	<i>IEC 60534-8-4 (2015)</i>
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Noise prediction data

Low-noise design		<i>n.a.</i>	
Sound velocity	cF1	<i>1,569.4</i>	m/s
Pressure level	LA1	<i>184.0</i>	dB
Pressure level	LA2	<i>8.31</i>	dB
Sound pressure level	Lpi	<i>193.0</i>	dB





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Noise prediction data (continued)

Transmission loss	TL	-78.5	dB
Sound power level (A-weighted)	LwAe	120.0	dB(A)
Sound pressure level of valve (A-weighted)	LpAe	109.0	dB(A)

Comments:

Reliability index - Ri

Sound pressure level: 70 %; High velocity: 29 %
Recommended solution: Heavy duty valve, Hardened seat/plug, Low-noise

Legend

- Errors
- Alarm

